

COURSE 2B - DA-to-DE Bridge

B04 Protected Practice Review

Open one review only after the matching genuine attempt is saved

Fast-track purpose

This review diagnoses reasoning. It does not replace your work. After reading the matching review, close it and complete the changed retry without copying wording.

Locked learning loop

Why -> picture -> new words -> watch -> analyst-to-engineer transition -> follow once -> observe -> explain -> mistakes -> recover -> fresh task -> validate -> teach back -> protected review -> changed retry -> evidence/route.

- not Course 2B FINAL

Review B04-L01 - Operational vs analytical workloads: OLTP and OLAP

Unlock condition: saved attempt exists in practice sheet Workload_Contract. If not, close this review now.

What strong work demonstrates

Correctly separates event-recording workload from historical decision workload and names source-system risk.

Common weak patterns

Calling every SQL database a warehouse; running dashboard-style full scans directly on the app database; treating OLTP/OLAP as vendor brands.

Targeted repair

Ask: is this action recording/changing one business event now, or summarizing many events to make a decision? Then identify who is harmed if the query is slow.

Changed retry - review closed

Fresh domain: clinic appointments, payments, weekly no-show analysis and annual capacity planning.

Evidence check

Every row has workload type + reason + source-risk note. At least one analytical request has an explicit "do not run directly on operational source" boundary. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

Review B04-L02 - Warehouse, lake and lakehouse mental models

Unlock condition: saved attempt exists in practice sheet Storage_Fit. If not, close this review now.

What strong work demonstrates

Chooses storage pattern from data shape, consumers, governance, access and replay needs, with a defensible rejected alternative.

Common weak patterns

Calling a folder of CSVs a governed lakehouse; assuming a lake is automatically cheaper overall; choosing a warehouse for raw binary/unstructured archives without a reason.

Targeted repair

Write four questions: What data shape? Who queries it? How much governance/transaction reliability? Must raw history remain cheaply replayable?

Changed retry - review closed

Fresh domain: e-commerce raw images/events, governed finance metrics and ML feature history.

Evidence check

Every decision names at least three requirements and one trade-off; one rejected alternative is defended. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

Review B04-L03 - Business keys, surrogate keys and SCD history

Unlock condition: saved attempt exists in practice sheet History_Keys. If not, close this review now.

What strong work demonstrates

Separates business identity from warehouse version identity and prevents ambiguous historical intervals.

Common weak patterns

Using a business key as if it uniquely identifies every historical version; Type 2 for every typo; overlapping effective intervals; silently deleting duplicate change rows.

Targeted repair

Ask whether the old value must remain historically true. If yes, history likely matters. Then state the source identity separately from the warehouse version identity.

Changed retry - review closed

Fresh domain: provider clinic change, spelling correction and specialty-history requirement.

Evidence check

Business key stated; history choice justified per attribute; no ambiguous active interval; duplicate change trap is surfaced rather than hidden. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

Review B04-L04 - ETL vs ELT and transformation boundaries

Unlock condition: saved attempt exists in practice sheet Transform_Boundaries. If not, close this review now.

What strong work demonstrates

Names the durable landing point and transformation boundary and can explain what is replayable after a bad rule.

Common weak patterns

Saying ELT means "no cleaning"; calling every sequence ETL; transforming raw evidence in place; assuming cloud always means ELT.

Targeted repair

Mark the first durable copy. Then ask: if tomorrow's rule changes, what can I replay without requesting the source again?

Changed retry - review closed

Fresh domain: subscription billing with raw landing and a sensitive-field pre-load rule.

Evidence check

Flow has explicit source, durable point, transform boundary, curated output and replay statement. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

Review B04-L05 - Layered analytical systems: raw, staged, conformed and serving

Unlock condition: saved attempt exists in practice sheet Layer_Contracts. If not, close this review now.

What strong work demonstrates

Defines layer contracts by allowed state, transformations, consumers and promotion quality gates rather than color labels alone.

Common weak patterns

Editing raw source rows; putting business KPIs into raw; using layer names with no consumer/quality contract; assuming every project needs exactly three physical layers.

Targeted repair

For each layer answer: what can enter, what may change, who trusts it, and what blocks promotion?

Changed retry - review closed

Fresh domain: logistics source events -> normalized shipments -> governed SLA mart.

Evidence check

Every layer has allowed state, transformation boundary, consumer, quality gate and recoverability statement. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

Review B04-L06 - Batch, micro-batch and streaming trade-offs

Unlock condition: saved attempt exists in practice sheet Latency_Plan. If not, close this review now.

What strong work demonstrates

Starts from freshness/recovery SLA and chooses batch/micro-batch/streaming with late-data and replay reasoning.

Common weak patterns

Choosing streaming because it sounds advanced; confusing refresh frequency with source event time; ignoring late data; promising zero latency.

Targeted repair

Start with "How stale can this safely be?" Then ask what happens if processing stops for 20 minutes.

Changed retry - review closed

Fresh domain: hospital telemetry alert, hourly utilization and monthly finance close.

Evidence check

Each mode has SLA + late-data + replay + cost/complexity rationale. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

Review B04-L07 - Lineage, metadata, ownership and observability basics

Unlock condition: saved attempt exists in practice sheet Lineage_Ownership. If not, close this review now.

What strong work demonstrates

Provides traceable upstream/downstream lineage, run metadata, ownership and failure-impact routing.

Common weak patterns

Treating a folder path as complete lineage; collecting logs with no run ID; naming an owner with no responsibility boundary; tracking only failures and not normal row/freshness evidence.

Targeted repair

Start from the wrong output and walk upstream one transformation at a time. Write the minimum evidence you would need at each hop.

Changed retry - review closed

Fresh domain: support tickets -> clean events -> SLA metrics -> executive dashboard.

Evidence check

Chain has upstream/downstream links, owner, freshness expectation, run ID evidence and one failure-impact route. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

Review B04-L08 - Data-quality contracts: block, warn and reconcile

Unlock condition: saved attempt exists in practice sheet Quality_Contract. If not, close this review now.

What strong work demonstrates

Connects each quality rule to severity, action, owner and evidence; reconciliation is independent and issues are not hidden.

Common weak patterns

Replacing errors with zero; deleting orphans to make counts match; marking every issue BLOCK; using one grand total as proof of all correctness.

Targeted repair

For each defect ask: can a consumer make a materially wrong decision if this passes? If yes, block or isolate until explicitly resolved.

Changed retry - review closed

Fresh domain: marketplace duplicate seller history + orphan payout + GMV reconciliation.

Evidence check

At least one BLOCK, one WARN/QUARANTINE, one independent reconciliation and named owner. No issue is hidden by deletion. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

Review B04-L09 - Replay, backfill and warehouse failure-recovery reasoning

Unlock condition: saved attempt exists in practice sheet Recovery_Runbook. If not, close this review now.

What strong work demonstrates

Preserves raw evidence, bounds recovery scope, protects state, prevents duplicates and proves post-recovery correctness.

Common weak patterns

Deleting bad output before preserving evidence; advancing checkpoint before reconciliation; rerunning unbounded history; validating only the metric that originally failed.

Targeted repair

Write seven verbs in order: detect, contain, identify scope, recover from preserved input, validate independently, promote, document.

Changed retry - review closed

Fresh domain: two missing support-event days where downstream daily aggregates already exist.

Evidence check

Runbook names affected window, source evidence, state/checkpoint handling, duplicate prevention, at least two independent controls and promotion/rollback rule. Save the changed retry separately; do not overwrite the first attempt.

Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.